

v1.5.0

The goal of this documentation is to give a short demo and some information about the new features of the version 1.5.0. This document does not explain everything in detail. For a detailed explanation read the core documentation of the asset! Also note that not every change is included by this document. Mostly the new features and those changes are worth mentioning.

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BuildingElementMetaData has been removed

Almost all of the properties of the MetaData were redundant or not used at all. The MetaData script was completely removed from the assets and the used properties repositioned to the BuildingElement's interface.

Improved Persistency Feature

The persistence feature was outdated as well as the tools. From now it can save the BuildingParents too. Also from now it saves all of the properties of the BuildingElements, Buildings, BuildingParents.

There are the possibilities to save and load the BuildingElements one by one. The Building are able to save too with all of it's elements' data or the whole BuildingParent can be saved and loaded abc into the scene.

BuildingManager Blocked Input Interface

A new function is available on the ABS_IBuildingManagerExternalInterface that blocks the BuildingManager's inputInterface.

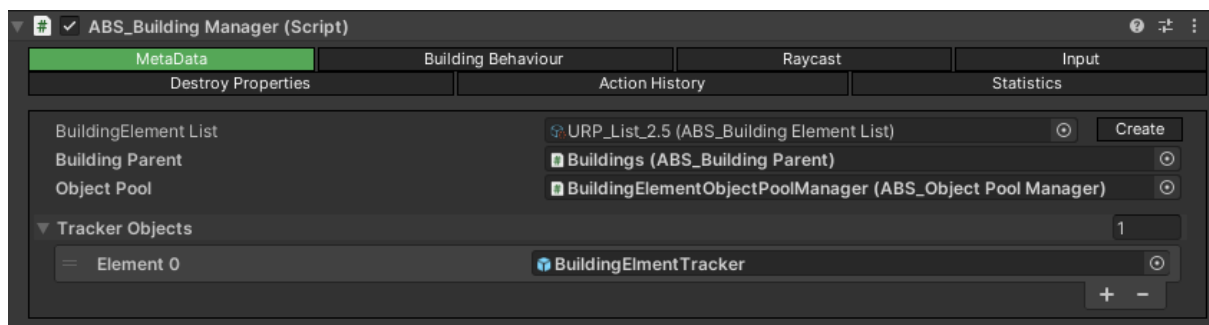
Note that it is not deactivating the manager it just blocks every input front he InputManager. This is a temporary blockage. With this function the manager can be unblocked as well.

```
/// <summary>
/// Block or unblock the input from the ABS_IBuildingManagerInputActionInterface on the BuildingManager.
/// </summary>
/// <param name="p_BlockState">Boolean value that represent the block state of the input interface.</param>
public void BlockInput(in bool p_BlockState);
```

Tab Based UI

Some script got a totally new UI design:

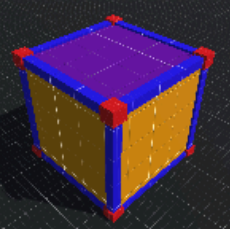
- Building Manager
- Building Element
- Building Algorithm Settings
- Building Loader (tool)
-



Improved Override Element Ruleset

In this version the Override Element Ruleset has been improved. From now the following new options are available:

- One To Set
 - The target element can override the elements from the set.
- Set to One
 - Any element from the set can override the target element.
- Both Way
 - The target element can override every element from the set.
 - Also every element can override the target element.
- Set
 - There will be no target element, just a set.
 - Every element can override any other element from the set.



OverrideElement Ruleset

Checkout

Name: Rule_Copy

Relation Type: One To Set

Override Target Object: no_Wall_2.5 (ABS_Building Element)

 The target element can be overridden by the following

Add element

BuildingElement Object: URP_AdvancedGridDemo_WallDoor_2.5 (ABS_Bui) **Delete**

Override Target Object :

8d53dd53-€		URP_AdvancedGridDemo_Wall_2.5 Prefab Guid: URP_AdvancedGridDemo_Wall_2.5
80c4e112-9		URP_AdvancedGridDemo_WallDoor_2.5 Prefab Guid: URP_AdvancedGridDemo_WallDoor_2.5

Save

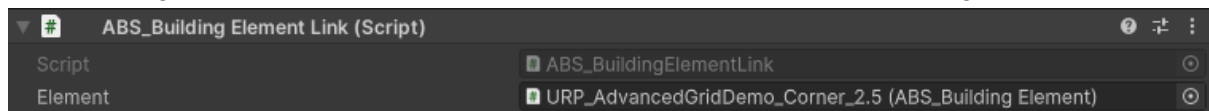
BuildingElement Link

There is the possibility when the Building Elements has a child object that has a collider that can be hit by the raycast. It can happen when the child object collider is bigger than the element's collider or exactly the same size.

This can cause problems during the destroy process because if it's raycast hits the child object's collider the algorithm will think that it does not hit any BuildingElement and it can break the destroy logic.

When the colliders are the same size then a "Z fight" light effect and some frames will hit the BuildingElement some will not.

This BuildingElement link is a script which holds a reference to the BuildingElement.

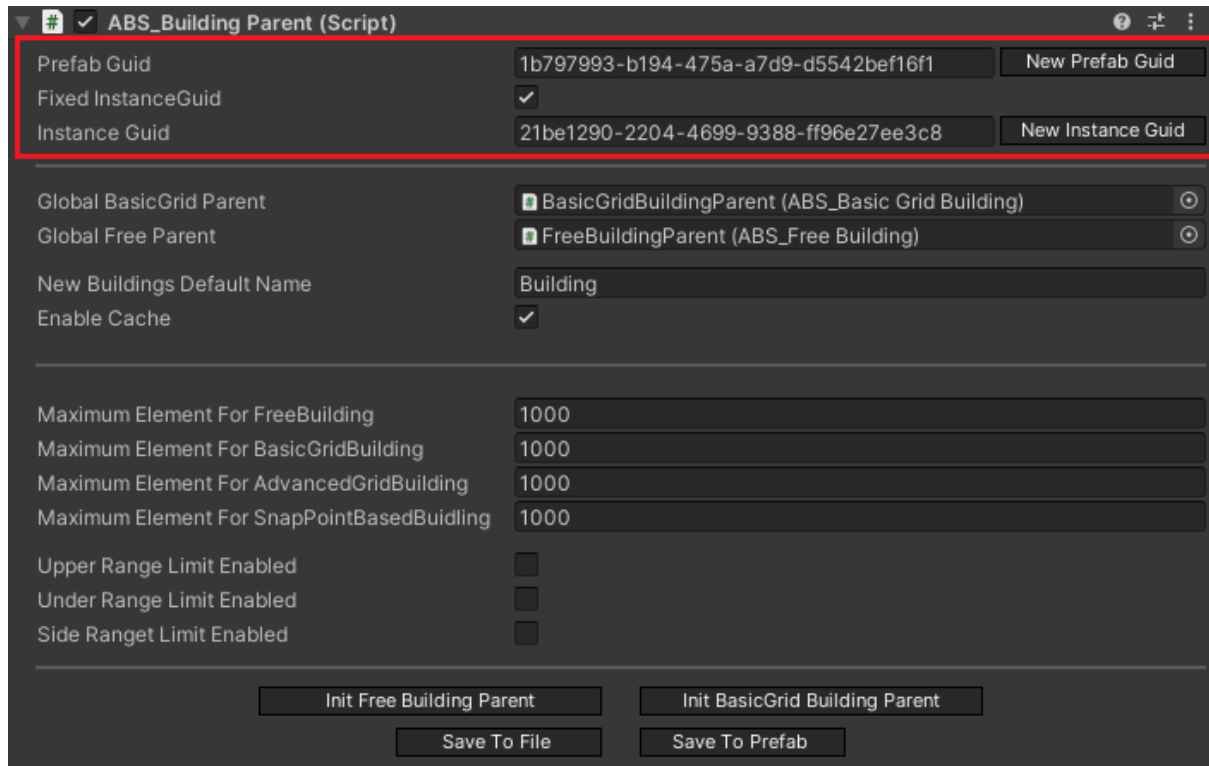


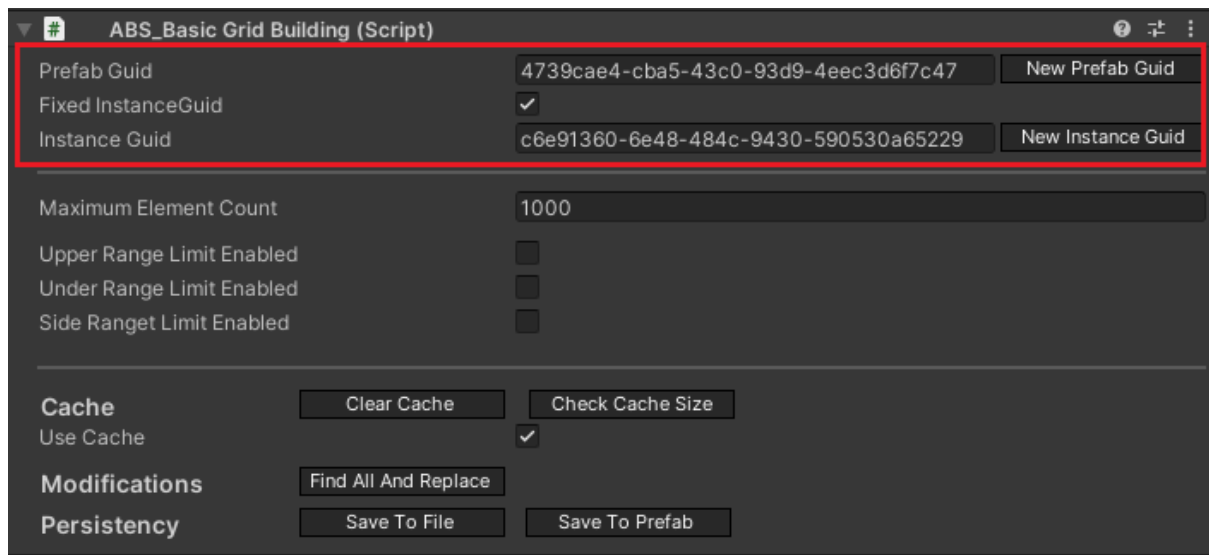
When the algorithm's raycast will hit a collider it will check if it has a BuildingElement or a BuildingElementLink script.

So the way you should use it is to add the link script to every child GameObject of the BuildingElement that has a collider and add the BuildingElement as reference to the links.

Guid changes

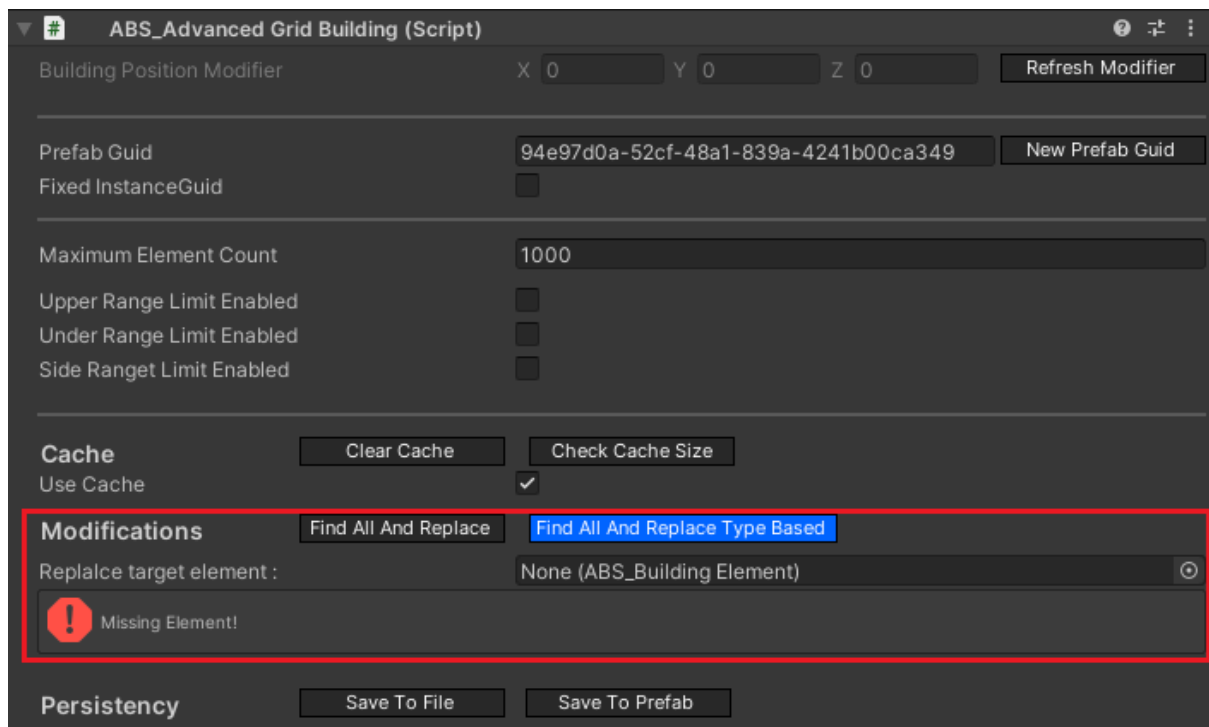
From now the Buildings and the Building Parents' UI are containing the prefab and the instance guid too.





New Modification feature for the AdvancedGridBuilding

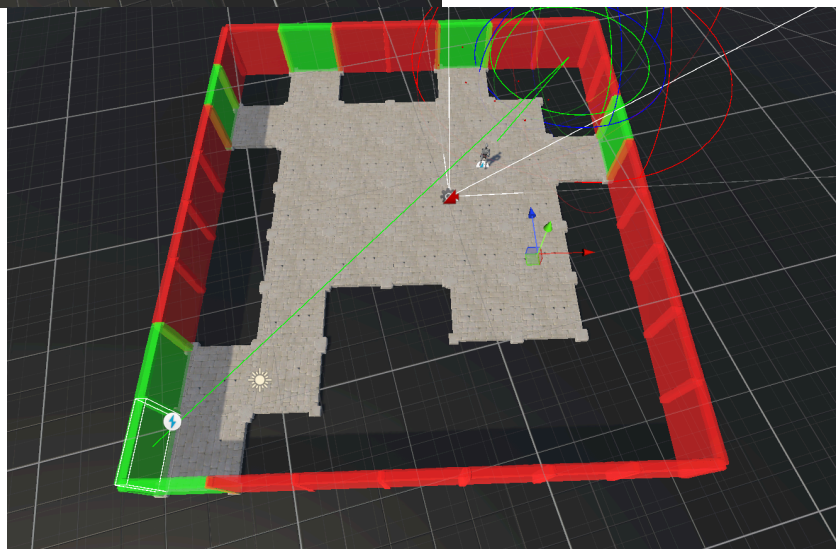
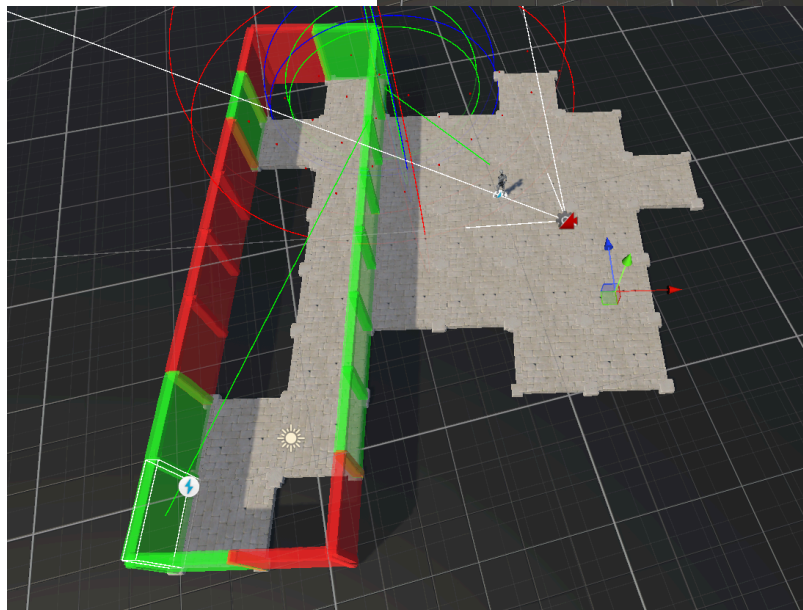
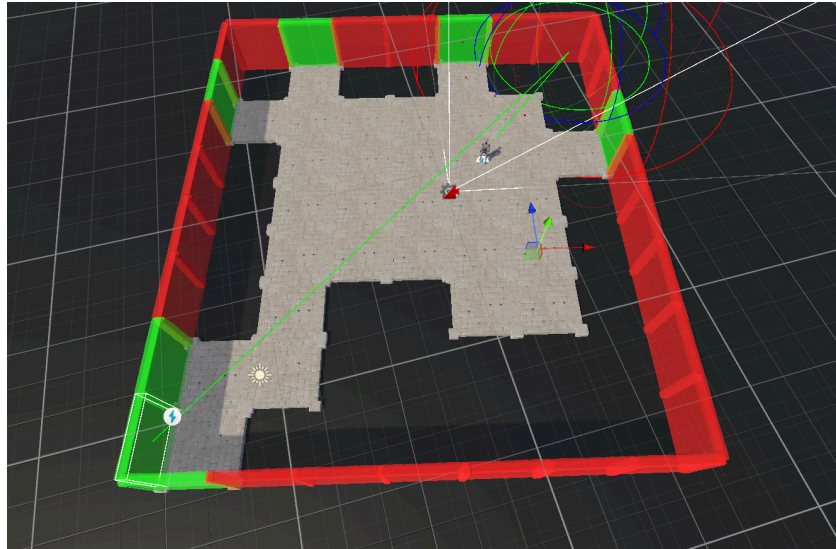
The AdvancedGridBuilding got a new type based find and replace feature. It will check every element of the building that has the same AdvancedGridType as the given element and replace them to that element.



AdvancedGridBuildingSnappointRuleSet

The 1.5 version introduced a new ruleset for the AdvancedGridBuilding that can be used to customize the algorithm snapping logic. With the rules snappoints can be blocked and denied for creating custom unique logics.

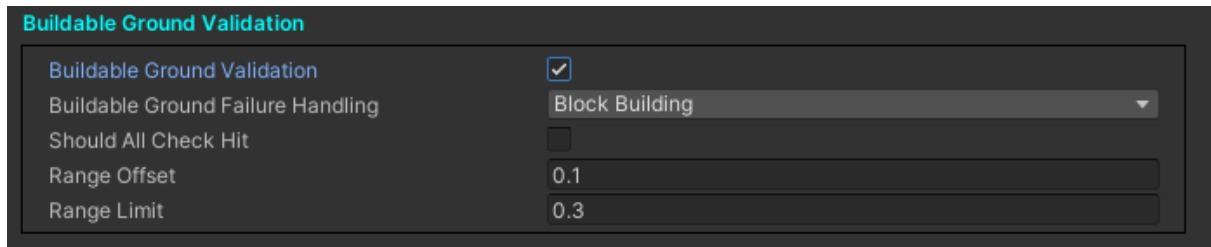
In the following example I blocked the snapping of the walls to each other so only those elements are validated in the following scenario where they are built on the top of the floor. Usually all of the walls would be validated in this scenario.



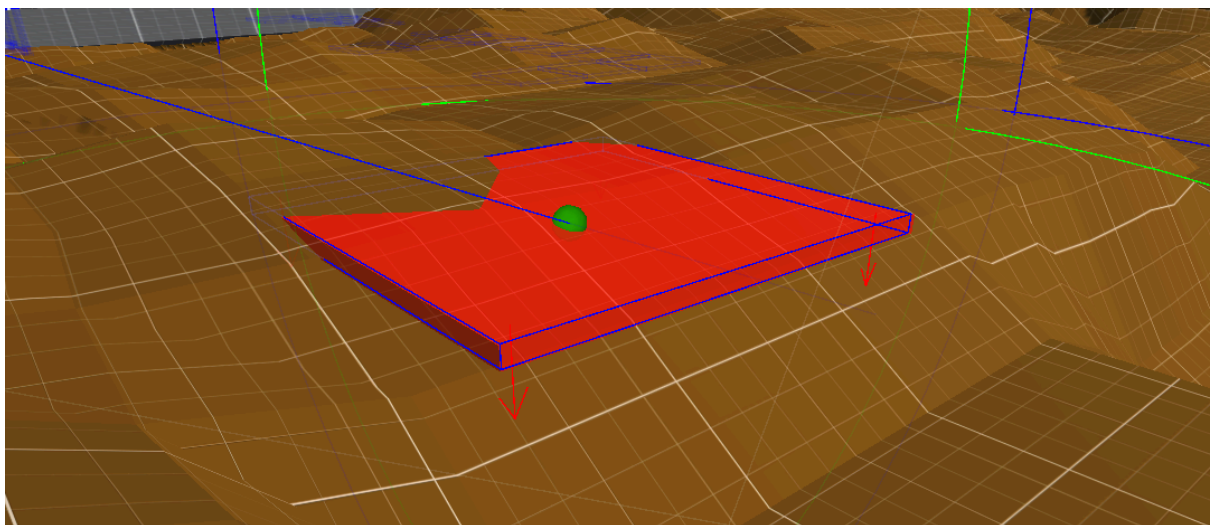
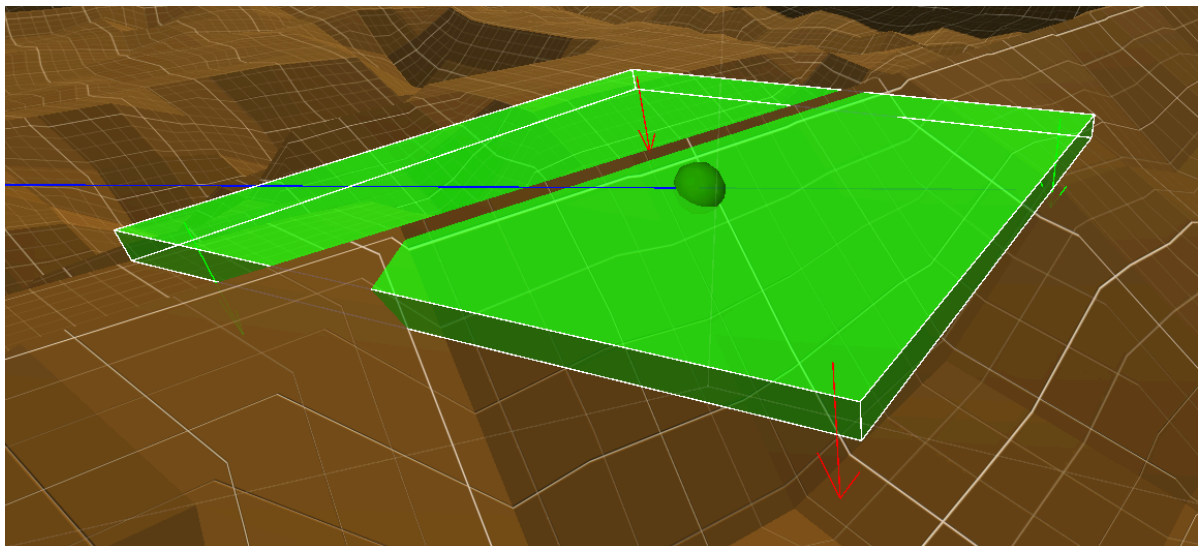
Buildable Ground Validation

From the version 1.5 a new position validation is available.

This validation tries to check if the ground is flat enough for building an element to that position.



This algorithm shoots raycasts down from the building element's bottom corners and counts how many raycast hit. If at least one of them hit the ground the position is validated.



New API error codes

SimpleBuildingProcessErrorCode

The tracker object has a new function:

```
public virtual void SimpleBuildResult(ABS_SimpleBuildingProcessErrorCode p_ErrorCode) { }
```

This function will be called at the end of a simple building process. The function will get a new error code as a parameter. The error code will contain the result of the simple building process.

```
namespace REST.AdvancedBuildSystem
{
    public enum ABS_SimpleBuildingProcessErrorCode
    {
        Unkown,
        Successful,
        HiddenElement,

        RuleBreak_AreaRules,
        RuleBreak_FoundationLogic,
        RuleBreak_ShouldOverride,
        RuleBreak_AllowBuildingInTheAir,
        RuleBreak_AboveMaximumElementCount,
        RuleBreak_PlayerCollision,

        //Building
        RuleBreak_Building_MaximumElementCount,
        RuleBreak_Building_InvalidPosition,
        RuleBreak_Building_UsedPosition,
        RuleBreak_Building_RangeRules,

        //Validation
        RuleBreak_Validation_UnderGround,
        RuleBreak_Validation_BuildableGround,
        RuleBreak_Validation_Collision,
        RuleBreak_Validation_ElementCollision,
        RuleBreak_Validation_AirHeightLimit,
        RuleBreak_Validation_Custom,

        //Specific AdvancedGridBuidling cases
        RuleBreak_AdvancedGridBuidling_ForbiddenAxis,
        RuleBreak_AdvancedGridBuidling_PositionRuleset,
        RuleBreak_AdvancedGridBuidling_PositionRuleset_Denied,

        //Specific FreeBuidling cases
        RuleBreak_FreeBuidling_RotationMaximumAngle
    }
}
```


ActionHistoryErrorCodes

The tracker object has 2 new function:

```
//-----  
//Action History  
public virtual void ActionHistoryRedoResult(ABS_ActionHistoryErrorCodes p_Result) { }  
public virtual void ActionHistoryUndoResult(ABS_ActionHistoryErrorCodes p_Result) { }
```

These functions will be called after the Redo or the Undo feature has been triggered. The purpose of this error code is giving information to the developers what happened during the history features.

```
public enum ABS_ActionHistoryErrorCodes  
{  
    Success,  
    Success_EmptyHistory,  
    Success_NoMoreAction,  
    Success_FeatureDisabled,  
  
    Failed_BuildingElementNotAvailable,  
    Failed_BuildingNotAvailable,  
    Failed_DeniedByTracker,  
  
    Failed_PositionValidation_UsedPosition,  
  
    Failed_DataValidation_UnkownGUID,  
    Failed_DataValidation_WrongParentObject,  
    Failed_DataValidation_WrongLocalPosition,  
    Failed_DataValidation_WrongLocalEulerAngles  
}
```

PersistencyLoadErrorCode

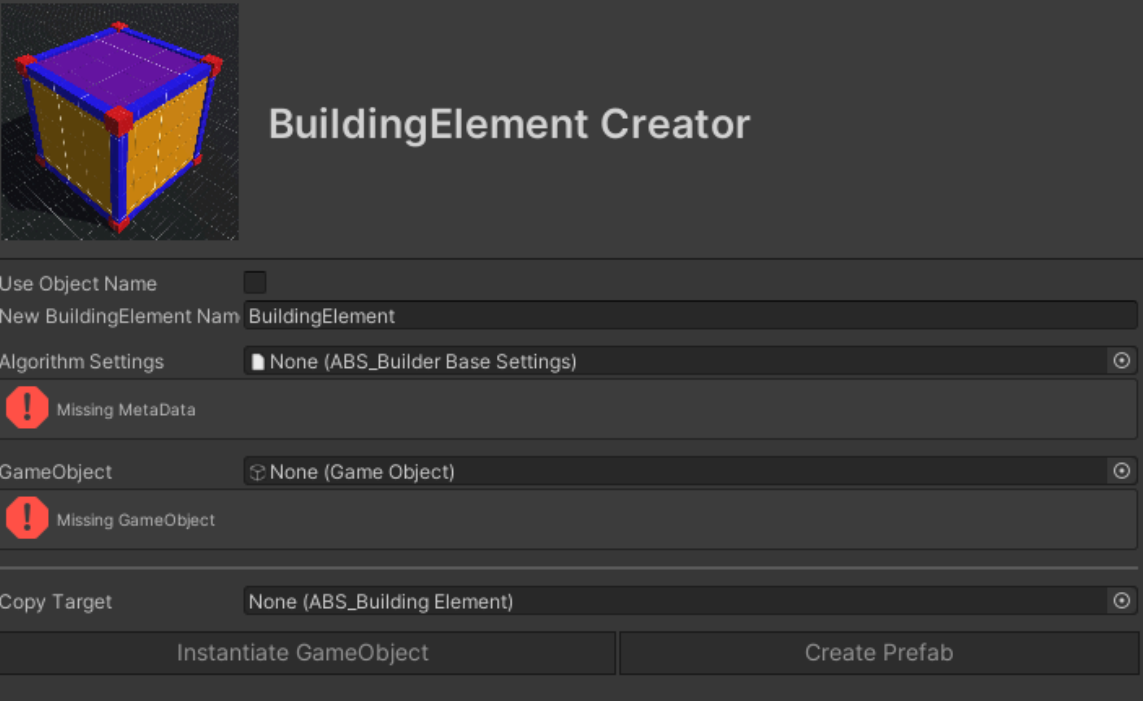
The new persistence load related functions are returning this new error code which will give information about the load process.

```
public enum ABS_PersistencyLoadErrorCode  
{  
    Unkown,  
    Successful,  
  
    JSON_NullOrEmptyString,  
    JSON_ErrorDuringLoad,  
  
    PersistedData_NullInput,  
    PersistedData_WrongDataType,  
  
    BuildingElementList_NullInput,  
    BuildingElementList_MissingElement,  
  
    Building_UnkownInstantiateError,  
    Building_AlreadyUsedPosition,  
  
    BuildingElement_UnkownInstantiateError,  
    BuildingElement_UnkownComponentError,  
    BuildingElement_NullPrefab,  
    BuildingElement_AddToBuildingFailed,  
  
    Update_WrongBasicObjectValues_BuildingParent,  
    Update_WrongBasicObjectValues_Building,  
    Update_WrongBasicObjectValues_BuildingElement,  
}
```

Tools

Building Element Creator

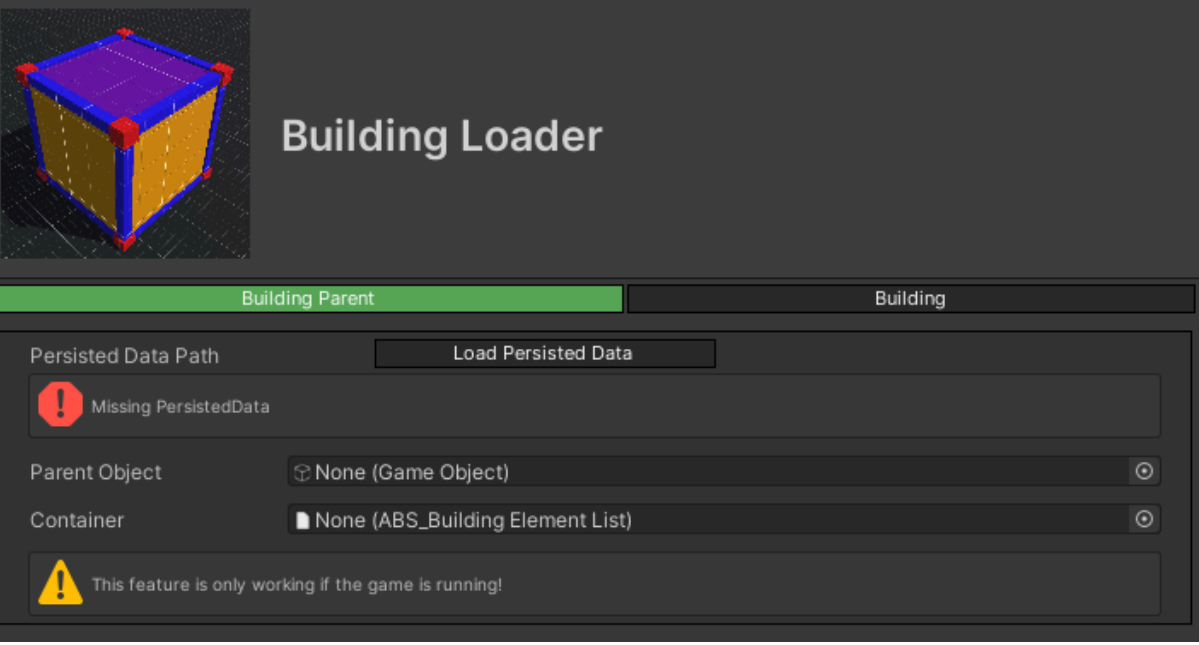
The Building Element Creator tool was outdated so in this update it had been updated. From now an optional copy target element can be added to what will be used during the creation process. Its properties will be copied to the new element.



The BuildingElement Creator interface features a 3D wireframe cube icon with red, yellow, and blue faces. The title "BuildingElement Creator" is displayed in a large, bold font. Below the icon, there are several input fields and buttons. The "Use Object Name" checkbox is unchecked. The "New BuildingElement Name" field contains the text "BuildingElement". The "Algorithm Settings" dropdown menu is set to "None (ABS_Builder Base Settings)". There are two red warning icons with exclamation marks, one labeled "Missing MetaData" and another labeled "Missing GameObject". The "GameObject" dropdown menu is set to "None (Game Object)". The "Copy Target" dropdown menu is set to "None (ABS_Building Element)". At the bottom, there are two buttons: "Instantiate GameObject" and "Create Prefab".

Building Loader

The Building Loader tool was outdated. This tool was updated too and from now the BuildingParents can be loaded as well.



The Building Loader interface features a 3D wireframe cube icon with red, yellow, and blue faces. The title "Building Loader" is displayed in a large, bold font. Below the icon, there are two tabs: "Building Parent" (selected) and "Building". The "Persisted Data Path" field is empty, and there is a "Load Persisted Data" button. There is a red warning icon with an exclamation mark labeled "Missing PersistedData". The "Parent Object" dropdown menu is set to "None (Game Object)". The "Container" dropdown menu is set to "None (ABS_Building Element List)". At the bottom, there is a yellow warning icon with an exclamation mark and the text "This feature is only working if the game is running!".

Scene Helper

The Scene Helper is a new tool that started with 1 feature, the Analyze. This feature analyzes the opening scene and checks every GameObject. It can recognise multiple kinds of errors of the scene and it can fix some of them as well. Currently it starts with 12 check categories that can recognise more than 12 kinds of errors such as missing properties or wrong parenting logic.

